Practical 5: Maven + Docker + CI/CD Integration

Aim

To containerize a web application using Docker and implement CI/CD using GitHub Actions.

Tools Required

Docker Desktop, Git, GitHub, Python (Flask)

Step 1: Create Project Folder

```
mkdir my-docker-app
cd my-docker-app
```

Step 2: Create Files

Create app.py, requirements.txt, Dockerfile

Step 3: app.py Code

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello from Docker CI/CD!"

if __name__ == "__main__":
    app.run(host='0.0.0.0', port=5000)
```

Step 4: requirements.txt

```
Flask
```

Step 5: Dockerfile

```
FROM python:3.9-slim
WORKDIR /app
COPY . .
```

```
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

Step 6: Build Docker Image

```
docker build -t my-docker-app .
```

Step 7: Run Docker Container

```
docker run -p 5000:5000 my-docker-app
```

Step 8: Output

Open http://localhost:5000
Output: Hello from Docker CI/CD!

Step 9: GitHub CI/CD Setup

```
mkdir .github
cd .github
mkdir workflows
cd workflows
Create ci-cd.yml file
```

Step 10: ci-cd.yml

```
name: CI-CD Pipeline

on:
  push:
    branches:
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Build Docker Image
        run: docker build -t my-docker-app .
```

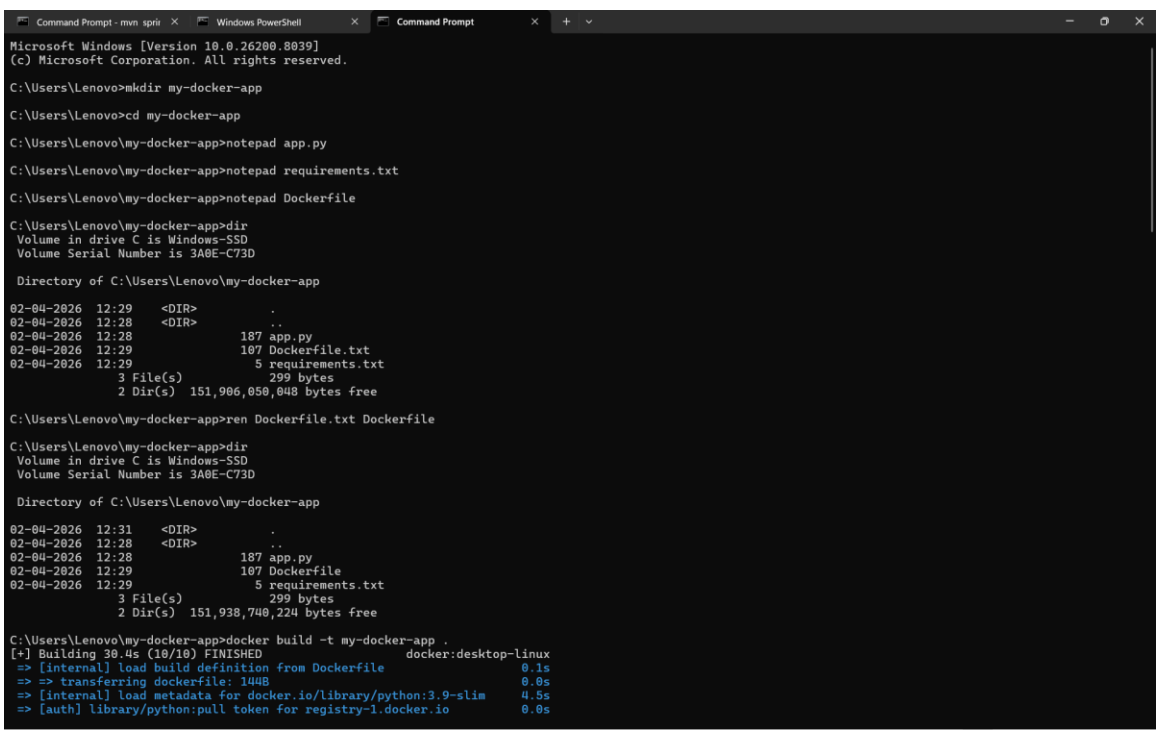
```
- name: List Docker Images
  run: docker images
```

Step 11: Push to GitHub

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin <repo_url>
git push -u origin main
```

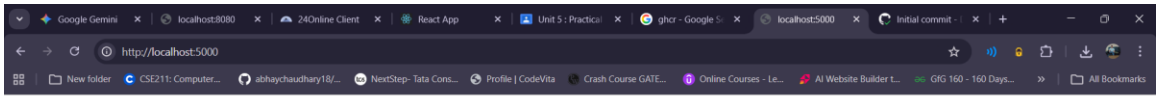
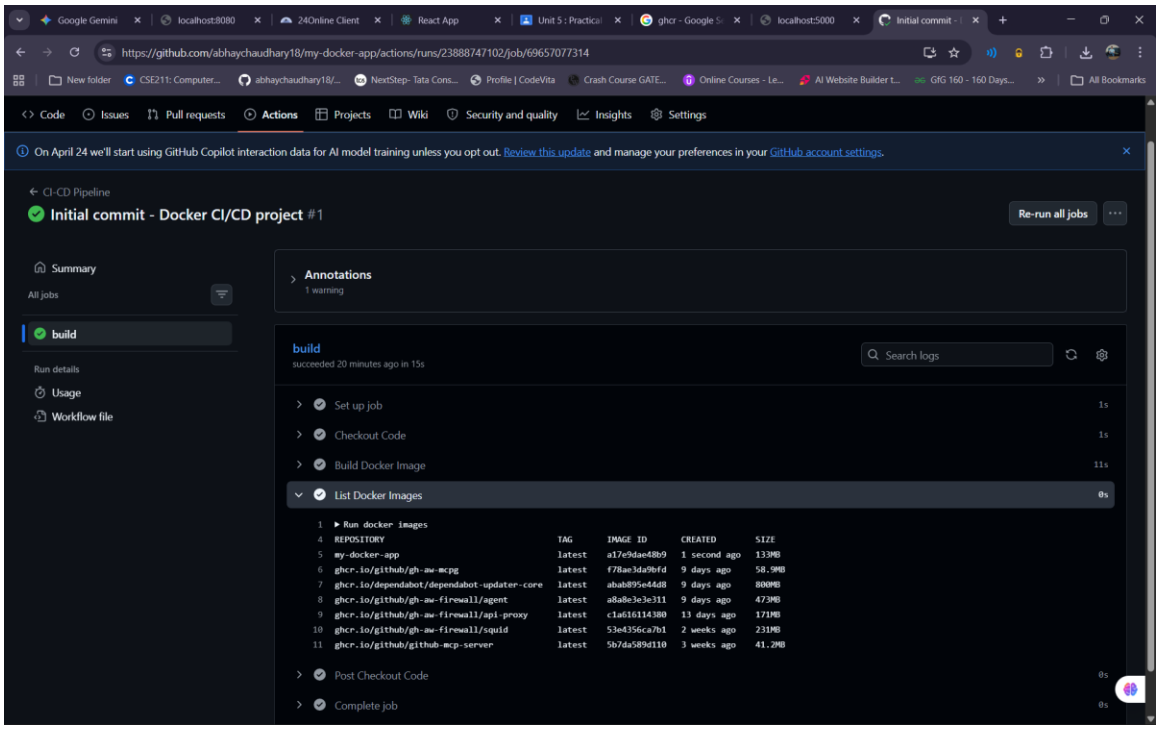
Conclusion

The application is successfully containerized using Docker and CI/CD pipeline is implemented using GitHub Actions.



```
Command Prompt - mvn sprir   Windows PowerShell   Command Prompt

C:\Users\Lenovo\my-docker-app>mkdir .github

C:\Users\Lenovo\my-docker-app>cd .github

C:\Users\Lenovo\my-docker-app\.github>mkdir workflows

C:\Users\Lenovo\my-docker-app\.github>cd workflows

C:\Users\Lenovo\my-docker-app\.github\workflows>notepad ci-cd.yml

C:\Users\Lenovo\my-docker-app\.github\workflows>cd ../../

C:\Users\Lenovo\my-docker-app>docker build -t my-docker-app
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information

C:\Users\Lenovo\my-docker-app>docker build -t my-docker-app .
[+] Building 10.3s (9/9) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.0s
 => => transferring dockerfile: 144B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                  1.4s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b            0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b            0.0s
 => [internal] load build context                                                                                                   0.1s
 => => transferring context: 550B                                                                                                   0.0s
 => CACHED [2/4] WORKDIR /app                                                                                                       0.0s
 => [3/4] COPY . .                                                                                                                  0.1s
 => [4/4] RUN pip install -r requirements.txt                                                                                       6.6s
 => exporting to image                                                                                                              1.7s
 => => exporting layers                                                                                                             1.0s
 => => exporting manifest sha256:032688374fd188f4a3e9f3c23c466140e0df2bfeb014fb0dbadfa1d9c07e18cb                                   0.0s
 => => exporting config sha256:6baef3109229b19e450ca0889b8a6e504a3e9929e4f1b16c4fc18a84b57e64f6                                     0.0s
 => => exporting attestation manifest sha256:5e18e665ece6a148d0846dc643331bdf0f183039d3afb528805af66d9efcddd5                       0.0s
 => => exporting manifest list sha256:0439d3d868b44c3832e2fc28563a0beecaa5f648064b1b9a1672ace1f3e78e87                              0.0s
 => => naming to docker.io/library/my-docker-app:latest                                                                             0.0s
 => => unpacking to docker.io/library/my-docker-app:latest                                                                          0.5s

C:\Users\Lenovo\my-docker-app>docker run -p 5000:5000 my-docker-app
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
```

```
Command Prompt - mvn sprir   Windows PowerShell   Command Prompt

 => => naming to docker.io/library/my-docker-app:latest                                                                             0.0s
 => => unpacking to docker.io/library/my-docker-app:latest                                                                          0.5s

C:\Users\Lenovo\my-docker-app>docker run -p 5000:5000 my-docker-app
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [02/Apr/2026 07:08:55] "GET / HTTP/1.1" 200 -

C:\Users\Lenovo\my-docker-app>git init
Initialized empty Git repository in C:/Users/Lenovo/my-docker-app/.git/

C:\Users\Lenovo\my-docker-app>git add .

C:\Users\Lenovo\my-docker-app>git commit -m "Initial commit - Docker CI/CD project"
[master (root-commit) 408dc62] Initial commit - Docker CI/CD project
 4 files changed, 35 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 requirements.txt

C:\Users\Lenovo\my-docker-app>git remote add origin https://github.com/abhaychaudhary18/my-docker-app.git

C:\Users\Lenovo\my-docker-app>git branch -M main

C:\Users\Lenovo\my-docker-app>git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 892 bytes | 178.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/abhaychaudhary18/my-docker-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Lenovo\my-docker-app>
```



Hello from Docker CI/CD!